

Assignment: Used Car Price Prediction Using Multiple Linear Regression

1. Problem Statement

A used-car marketplace wants to build a machine learning system that can estimate the selling price of a used vehicle based on its characteristics.

You are working as a **Junior Machine Learning Engineer**. Your task is to analyze historical used-car data and develop a **Multiple Linear Regression model** that predicts the price of a vehicle.

Your solution must not simply train a model and report its accuracy.

You must demonstrate your understanding of:

- Data exploration
- Data cleaning
- Exploratory Data Analysis
- Feature selection
- Categorical variable encoding
- Train-test splitting
- Multiple Linear Regression
- Regression evaluation metrics
- Residual analysis
- Model assumptions
- Multicollinearity
- Overfitting and underfitting
- Feature interpretation
- Model improvement
- Business interpretation

2. Dataset

Use the following Kaggle dataset:

Used Car Price Prediction Dataset

The dataset contains approximately 4,009 used-car records with features describing the vehicle and its condition.

Important attributes include:

- Brand & Model
- Model Year
- Mileage
- Fuel Type
- Engine Type
- Transmission
- Exterior Color
- Interior Color
- Accident History
- Clean Title
- Price

Dataset source:

<https://www.kaggle.com/datasets/taeefnajib/used-car-price-prediction-dataset>

3. Business Objective

The company wants to answer:

Given the characteristics of a used car, what should be its expected market price?

Your model should help a seller or buyer estimate a reasonable price.

Target Variable

price

Input Variables

You should investigate which available features can be used to predict price.

Do not blindly use every column.

Instructions:

- Perform EDA before model building.
- Perform Feature Engineering
- Perform Data Preprocessing
- Perform Train-Test Split
- Build the Multiple Linear Regression Model
- Model Evaluation
- Actual vs Predicted Values
- Residual Analysis

- Assumptions of Linear Regression
- Feature Coefficients

Notebook Quality Requirements

Your notebook must contain:

- Clear markdown explanations
- Meaningful variable names
- Properly labelled graphs
- Appropriate graph titles
- No unnecessary code
- No repeated code
- Comments wherever required
- Interpretation after important outputs
- No unexplained library functions
- No copied Kaggle notebook code

A notebook containing only code and outputs will receive reduced marks.